

ESConv Task

The [ESConv dataset](#) contains ~1.3K emotional-support conversations between a help-seeker and a trained supporter. Each supporter turn is annotated with one of 8 support strategies (e.g. Question, Reflection of Feelings, Providing Suggestions), alongside the seeker's problem type and emotion. The dataset is released as a single file; **you define the train/validation/test split and must justify it.**

Build a deep learning model that predicts the supporter's next strategy from the preceding dialogue context. Provide solutions for the following:

1. The full 8-class problem.
2. A 3-class problem over the coarse groupings: Exploration, Comforting, Action. State which of the 8 strategies you map into each class and why.

Both solutions should include description of the methods chosen, codebase, reporting of results using established classification metrics, and interpretability / explainability of the proposed model's behavior. Report per-class results, not only aggregates - the label distribution is heavily skewed and an aggregate number alone will not be accepted.

Your report must open with a **split protocol** section stating your grouping unit and reporting the measured performance gap between a random-utterance split and your chosen split. This number is a required result.

- Comparison to the results in the paper published with this dataset will be a plus.

Generative AI based subtask:

- For every supporter turn in your test set, use generative AI to generate the supporter's response, under two conditions: conditioned on your model's *predicted* strategy, and conditioned on the *gold* strategy.
- Your output should be an excel file with the following fields: conversation id, turn index, problem type, dialogue context, gold strategy, predicted strategy, generated response (max 40 words), judged quality score, judged strategy adherence.
- Quality and adherence are to be scored by an LLM judge whose prompt you design. You must then **annotate at least 30 of these items yourself** and report agreement between your annotation and the judge (κ or QWK). Low agreement is a legitimate finding; report it.
- Your implementation should use a generative AI model. You should experiment with multiple approaches including different prompting alternatives based on prompt engineering best practices.
- Please supply 2 output files for your two best alternatives.

Additionally required:

- A **failed experiments** section: at least two things you tried that did not work, the evidence, and your diagnosis of why.
- A short listing of **instances where an AI tool gave you something wrong** and how you caught it.
- An **error analysis of** ~10 test items you read by hand where the algorithms got it wrong. With details about the error vs the ground truth and your thoughts about the reasons.
- A **cost and latency table** for the generative component: tokens used, cost, and what you would change to run it at 100× scale.

Before you begin the task (mandatory)

- Take [this](#) short training.